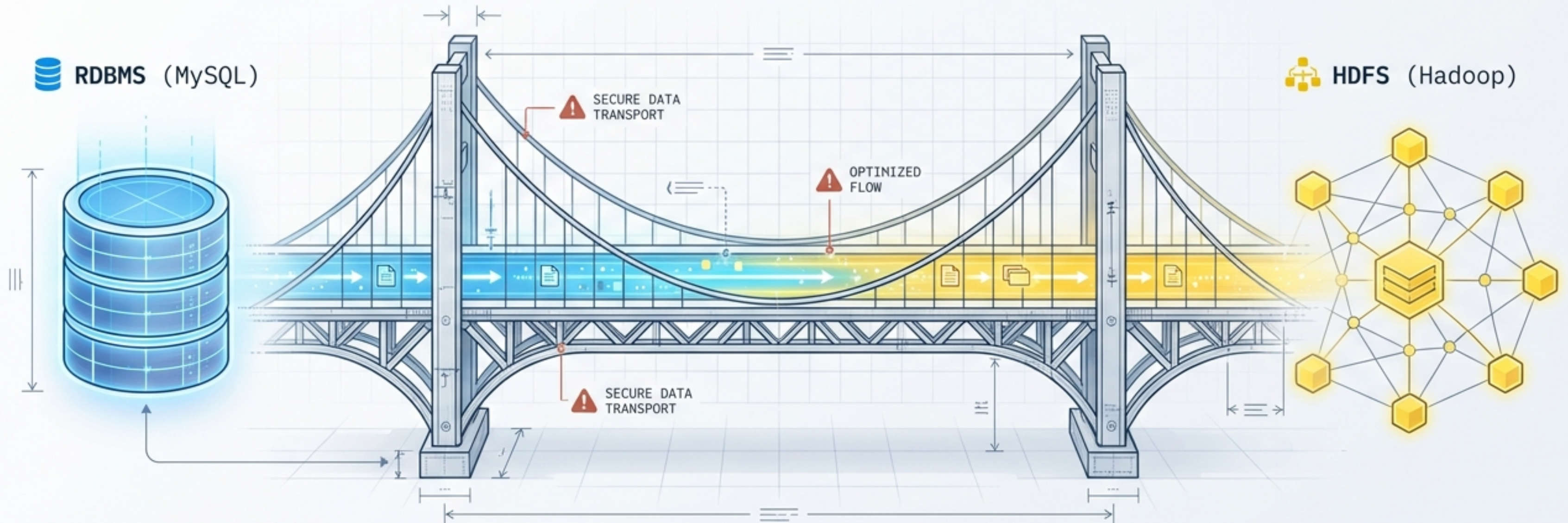
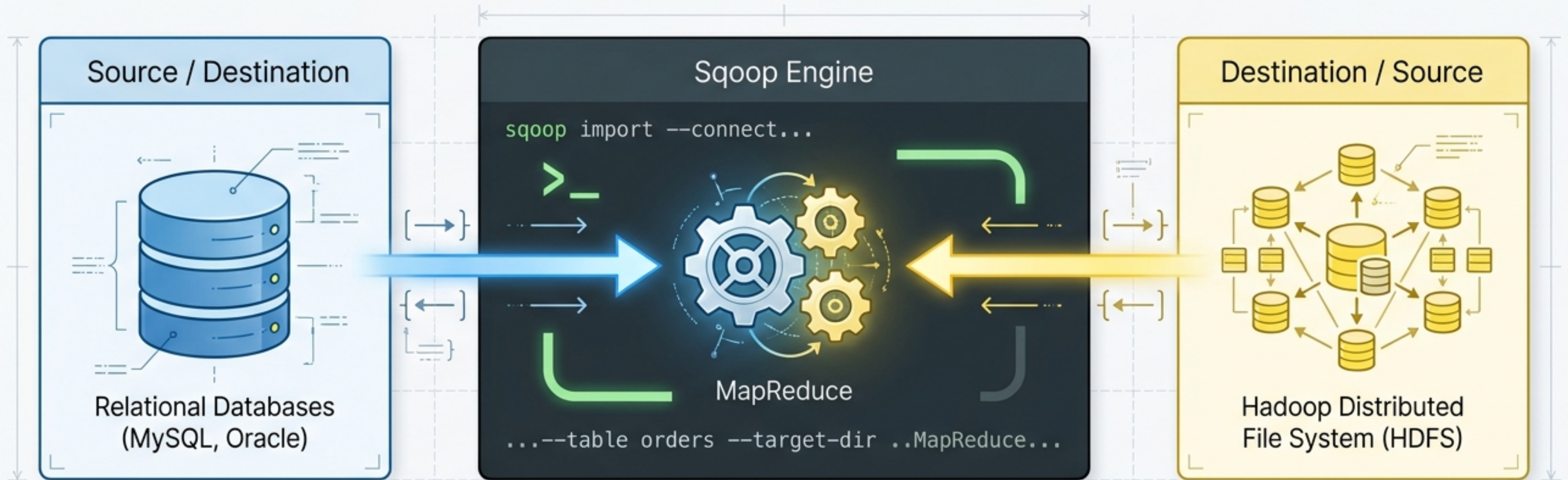


Apache Sqoop: Architecture & Operations

A Complete Study Guide for RDBMS to HDFS Data Migration



The Bridge Between Worlds: Sqoop's Core Architecture



Discover



List databases and map source tables.

Manipulate



Execute SQL directly via `eval`.

Ingest



Import data securely to HDFS.

Egest



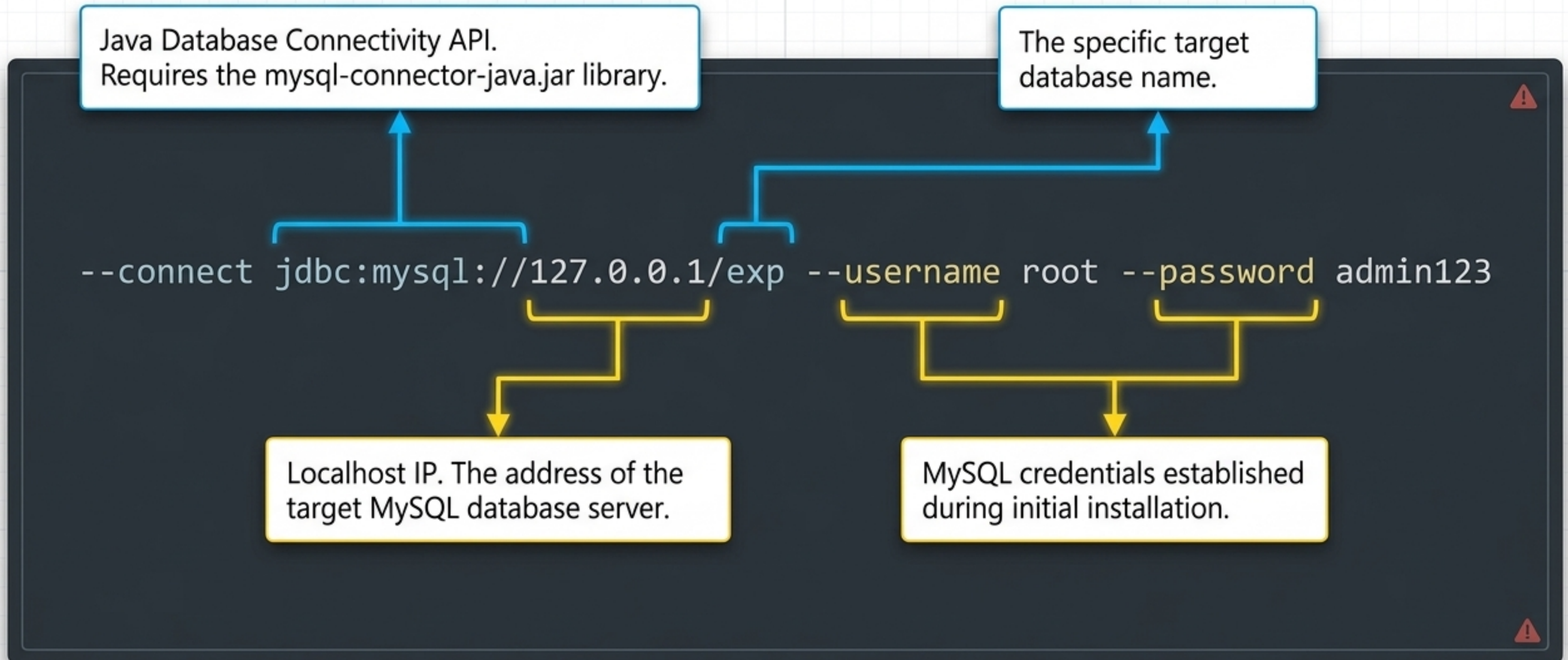
Export processed data back to RDBMS.

Automate



Schedule daily pipeline jobs.

The Universal Pre-Requisite: Establishing the JDBC Connection



Phase 1: Database & Table Discovery

Locating Databases

```
● ● ●  
$ sqoop list-databases --connect jdbc:mysql://127.0.0.1/  
...  
[OK] Databases found:  
test  
exp
```

Mapping Tables inside /exp

```
● ● ●  
$ sqoop list-tables --connect jdbc:mysql://127.0.0.1/exp  
...  
[OK] Tables found:  
emp  
test_new
```

Key Insight: Discovery commands run entirely from the Linux terminal, interrogating the live MySQL architecture in real-time.

Phase 2: Remote Database Manipulation (eval)



Syntax Spotlight

```
sqoop eval --connect jdbc:mysql://127.0.0.1/exp --query "CREATE TABLE test_new (id INT, name VARCHAR(20))"
sqoop eval --connect jdbc:mysql://127.0.0.1/exp --query "INSERT INTO test_new VALUES(1, 'swipe')"
```

Supported eval Operations

DDL CREATE, ALTER, DROP	DML INSERT, UPDATE, DELETE	Analytics SELECT, MAX, MIN, COUNT
Filtering WHERE clauses	Sorting ORDER BY, GROUP BY	Relational JOINS across tables

Phase 3: Data Ingestion via the **import** Command

```
sqoop import  
--connect  
  jdbc:mysql://127.0.0.1/exp  
--table test_new -m 1
```



HDFS automatically generates a destination directory matching the table name (e.g., /test_new). No manual creation is required.

The 5 Import Scenarios

☒ 1. Single Table (Baseline)

☐ 2. All Tables

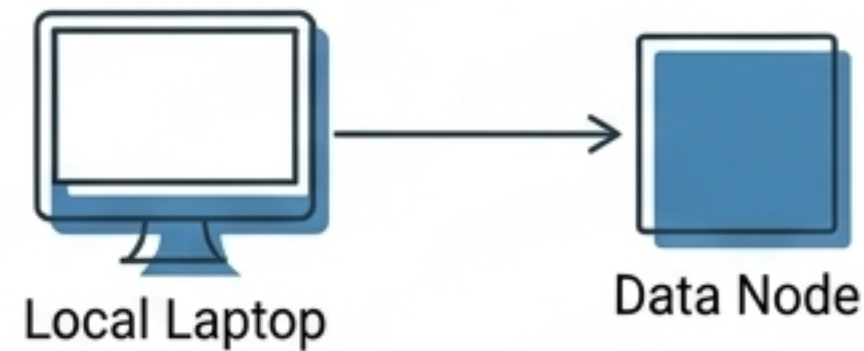
☐ 3. Target Directory

☐ 4. Conditional

☐ 5. Incremental Append

Optimizing Parallel Ingestion: Understanding Mappers (-m)

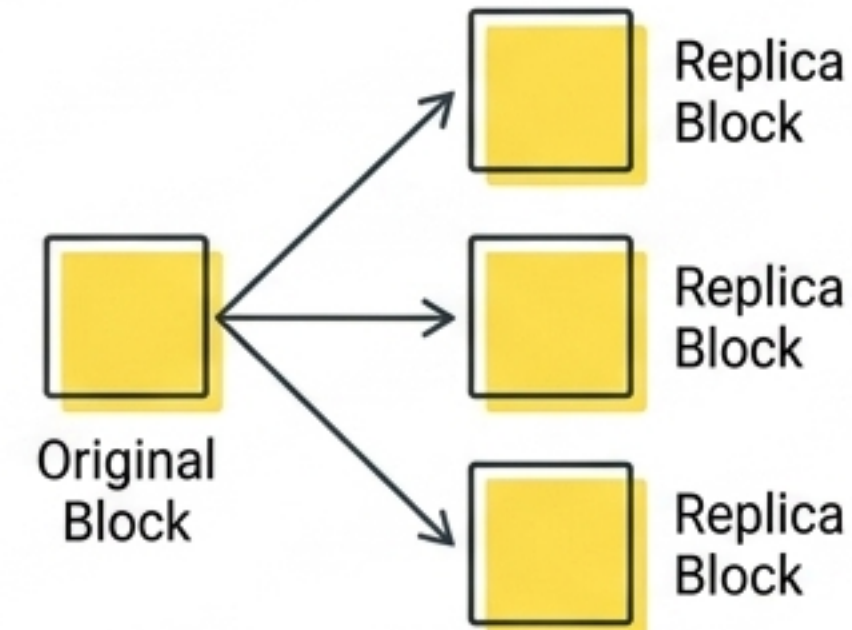
Local / Laptop Setup



`-m 1`

Used for testing. Maps directly to the single active data node on a local Ubuntu installation.
1 Node = 1 Input Split.

Production Environment



`-m 4`

The industry standard. MapReduce executes in parallel, utilizing multiple mapper tasks corresponding to the clustered data blocks.

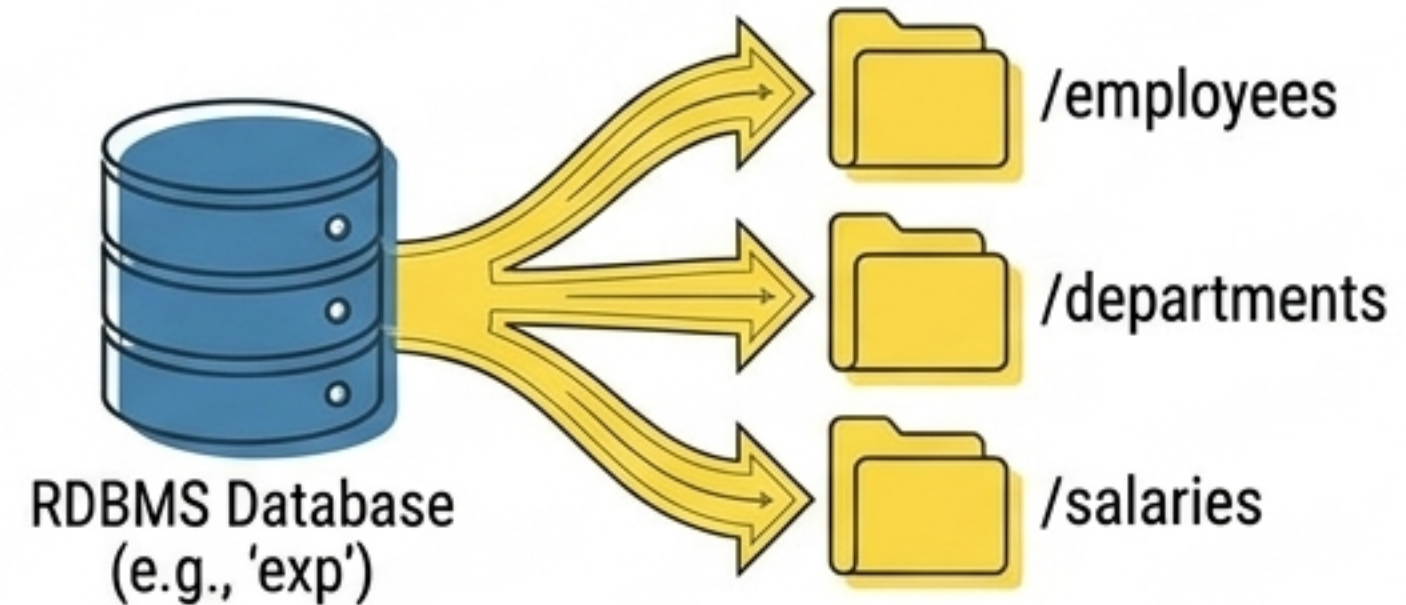
Advanced Ingestion Routing: Bulk & Custom Targets

Bulk Import Scenarios

```
sqoop import-all-tables  
--connect jdbc:mysql://127.0.0.1/exp  
--username root  
...
```

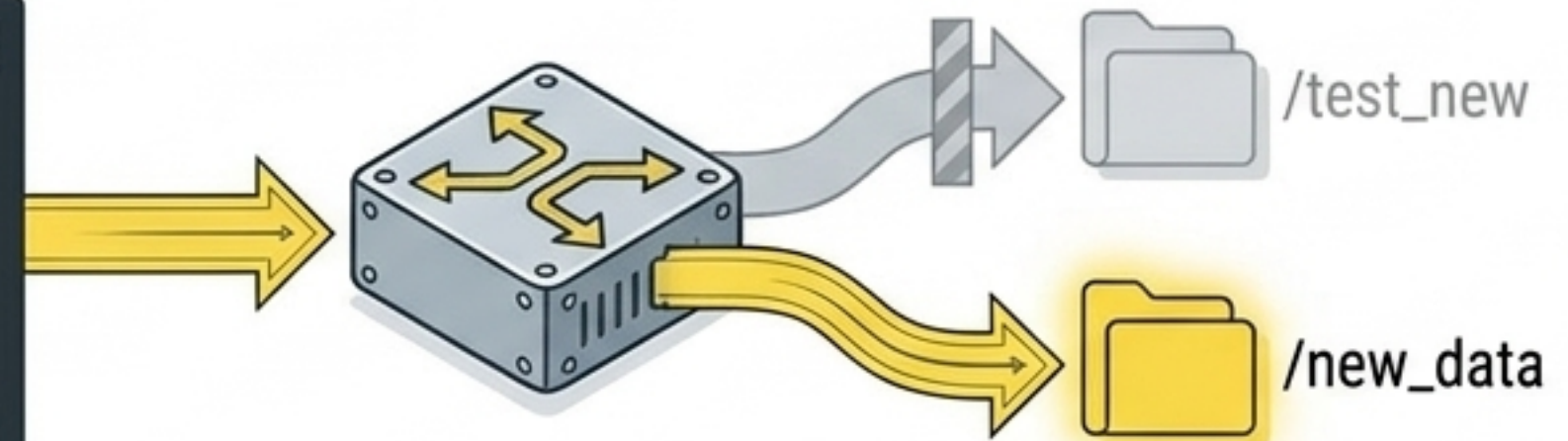


--table parameter
explicitly removed.



Custom Target Routing

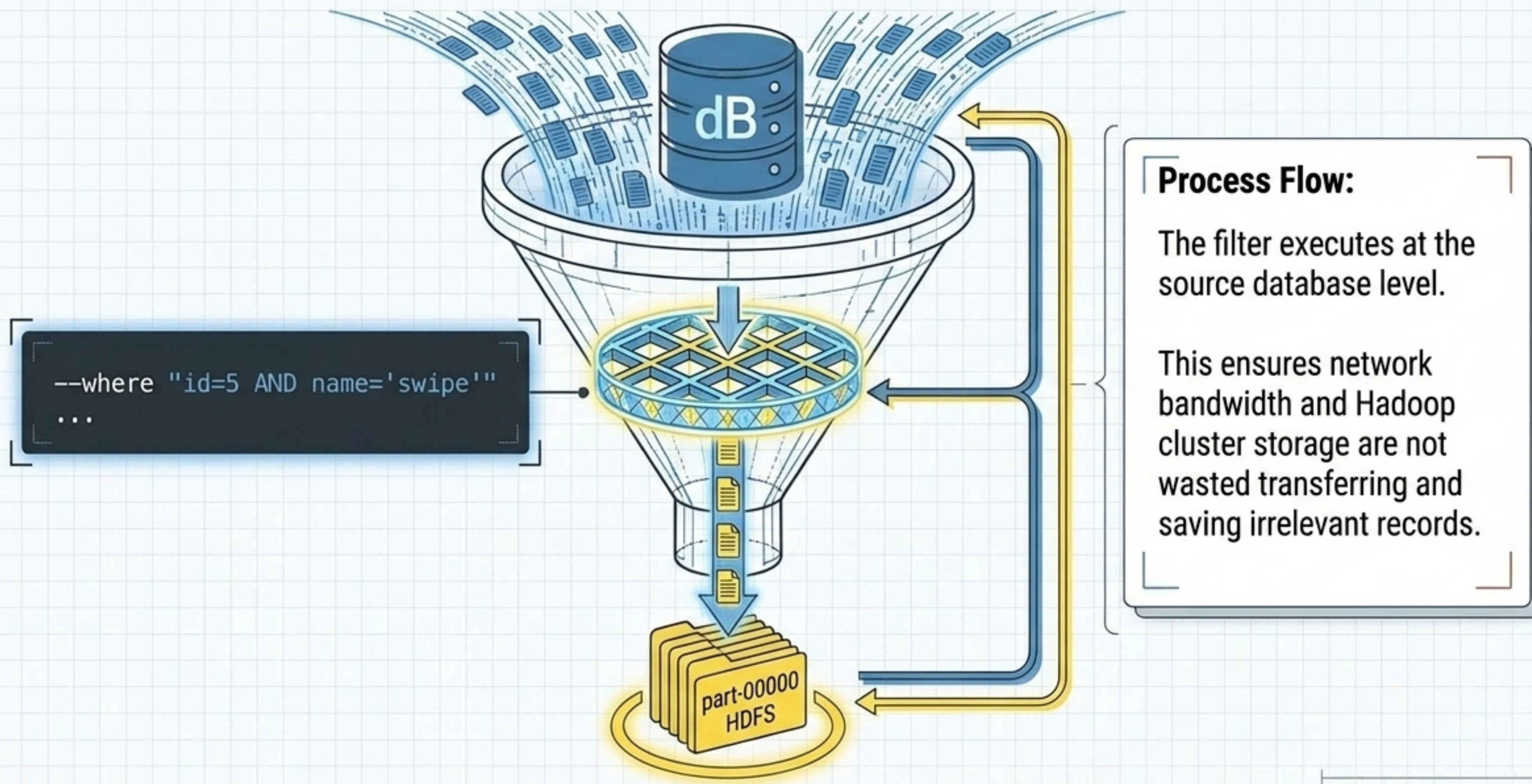
```
sqoop import  
--table test_new  
--target-dir /new_data  
--connect jdbc:mysql://127.0.0.1/exp  
...
```



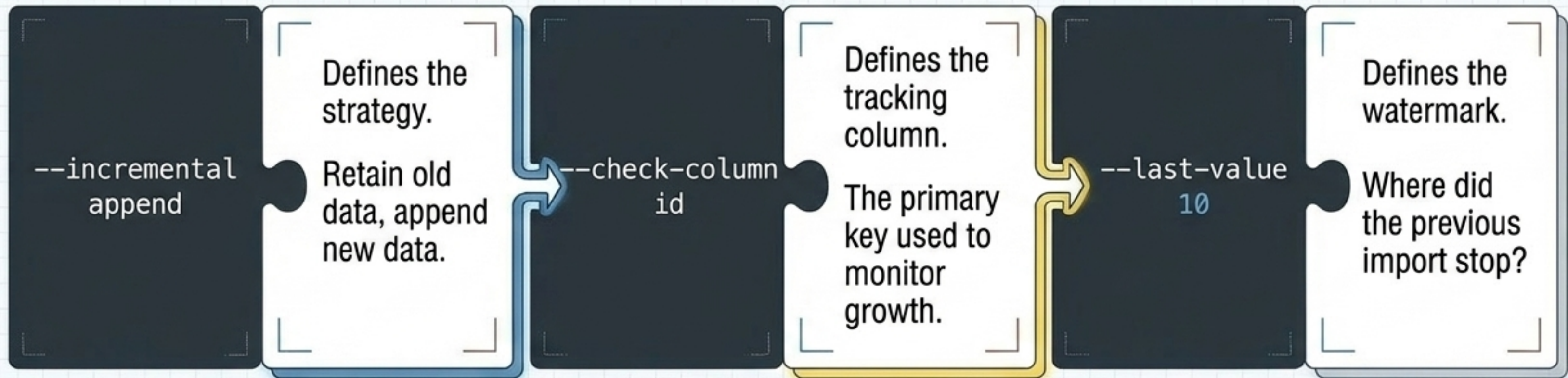
MapReduce Background Process:

Regardless of destination, MapReduce ultimately generates comma-separated text files (part files) inside these directories.

Precision Ingestion: Conditional Imports

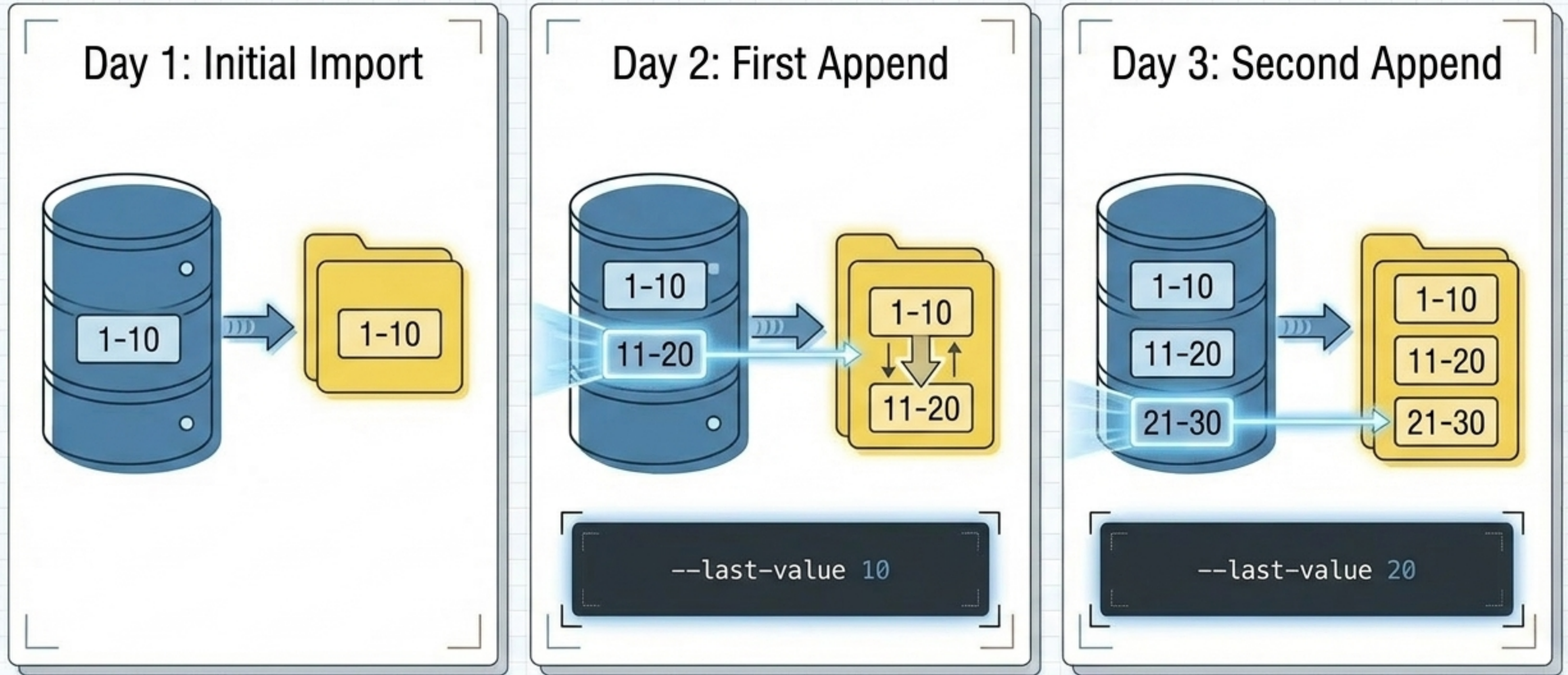


Sustaining the Pipeline: Incremental Append

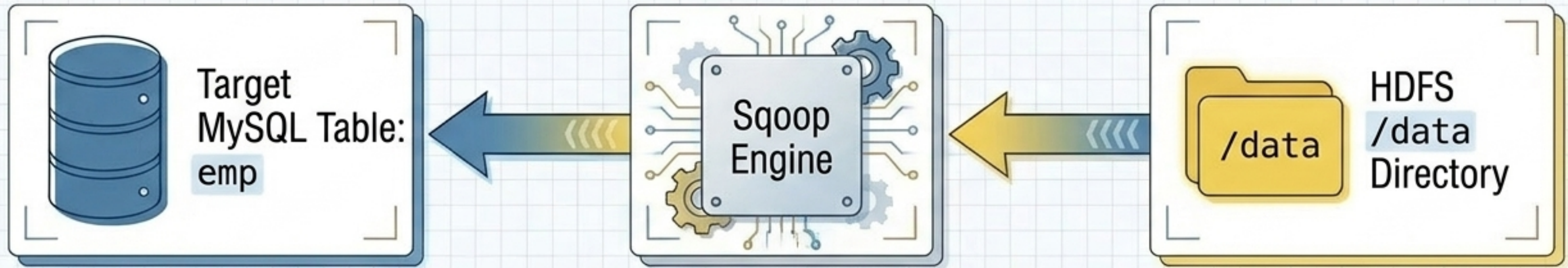


This triad of parameters prevents the costly re-ingestion of the entire table. It ensures Sqoop fetches only the net-new records added to the RDBMS since the previous execution.

Visualizing the Incremental Append Timeline



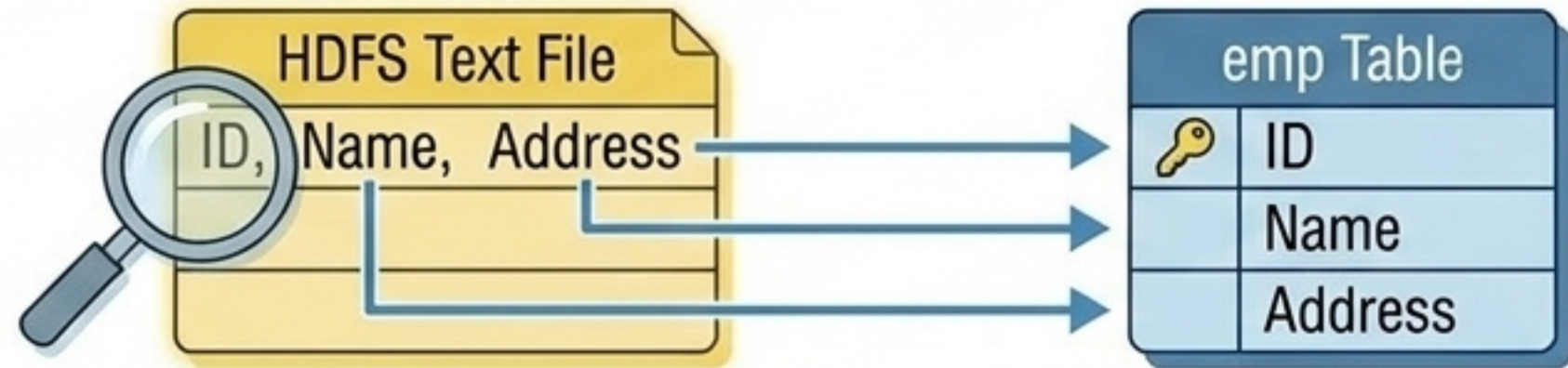
Phase 4: Data Egestion via the export Command



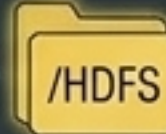
```
sqoop export --connect jdbc:mysql://127.0.0.1/exp  
--table emp --export-dir /data
```

Warning

**Strict Rule:
The Schema Must Match**



Diagnostic Matrix: Import vs. Export

	import  	export  
Core Routing Flag	<code>--target-dir</code>	<code>--export-dir</code>
Schema Pre-requisite	Auto-creates target HDFS directory based on table name.	 Requires pre-existing, exactly matching MySQL table schema. 
Output Format	Generates comma-separated "part" text files inside HDFS. 	Populates structured SQL rows inside RDBMS. 
Primary Use Case	 Archiving transactional data into the data lake. 	Pushing processed Big Data analytics back to end-users. 

Phase 5: Pipeline Automation (job)

1. Create

```
sqoop job --create my_job -- import...
```

Wraps any complex command (list, import, export) into a saved, reusable job template.

2. List

```
sqoop job --list
```

Displays all saved jobs currently residing in the environment.

3. Show

```
sqoop job --show my_job
```

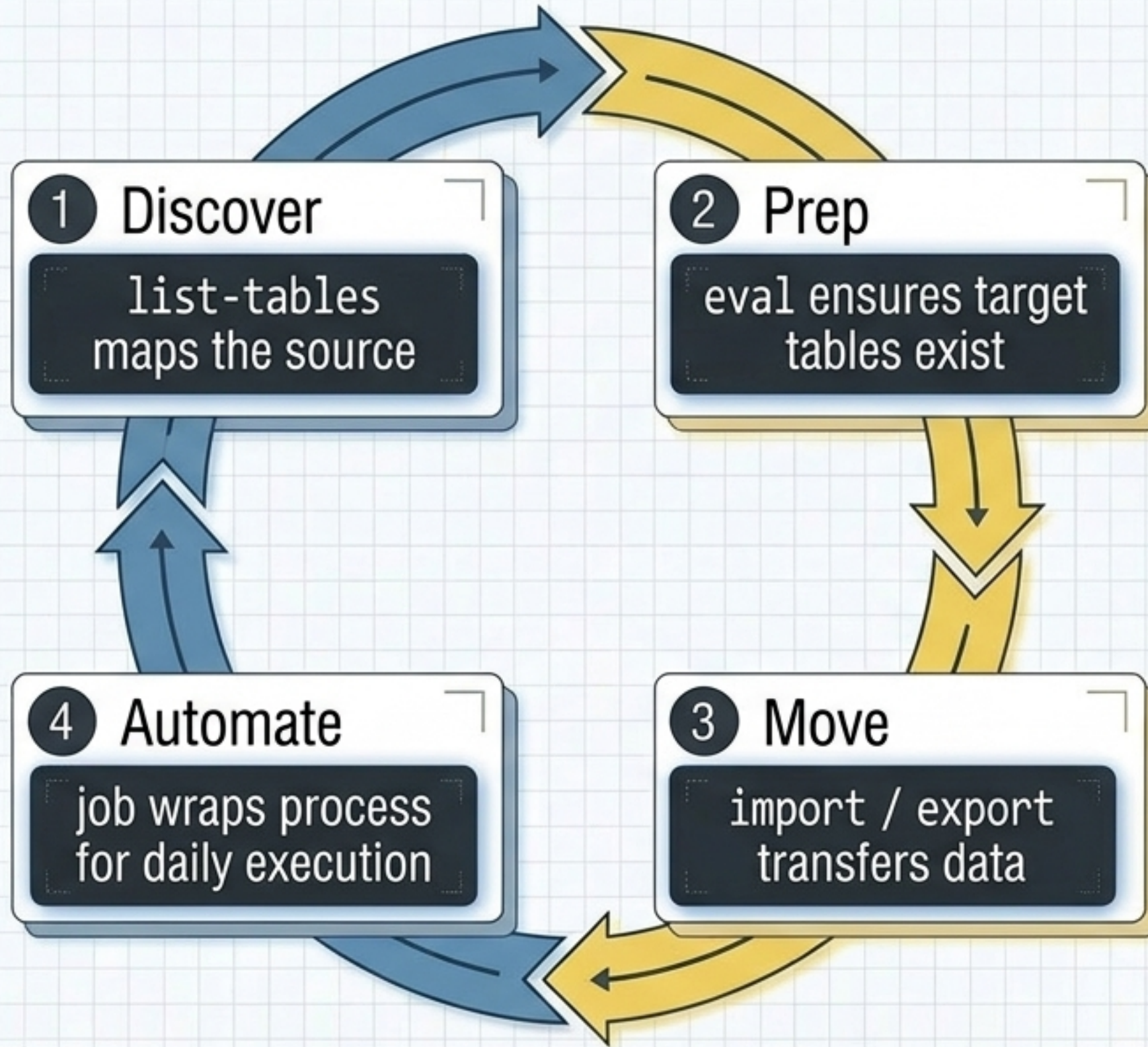
Inspects the hidden metadata, parameters, and execution plan of the specific job without running it.

4. Execute

```
sqoop job --exec my_job
```

Fires the automated task. Eliminates daily manual terminal typing.

Synthesis: The Complete Sqoop Workflow Pipeline



Installation & Pre-requisites

- ☐ Install MySQL (Configure root / admin123 credentials).
- ☐ Install Apache Hadoop (Single Node).
- ☐ Install Apache Sqoop.
- ☐ Place `mysql-connector-java.jar` in Sqoop library directory.
- ☐ Launch Ubuntu Linux terminal to begin testing.